CREATIVE / SOFTWARE

LIVE

Generative Poster System

Parametric design tool that produces seeded, reproducible layouts from a grammar of grids, palettes, and type scales. Exports print-ready SVG and PDF.

`p5.js` `Canvas` `SVG` `TypeScript`

VERSION	STATUS	OUTPUT	RUNTIME	OWNER	UPDATED
1.6.0	Live	SVG, PDF/X-4	Browser	Studio Tools	Sep 2026

1. Overview

The Generative Poster System composes posters from a written grammar rather than from a canvas of hand-placed elements. A design is declared as a set of grid rules, palette constraints, and a type scale; the system samples from that space using a seeded random number generator and lays out the result. Changing the seed produces a different poster in the same family, and re-entering an old seed reproduces that poster exactly.

Reproducibility is the whole point. Every output carries the grammar version and seed in its metadata, so a poster approved in a review can be regenerated at print resolution months later without keeping the file. Export goes straight to print-ready artwork: vector SVG for digital use and PDF/X-4 with bleed, crop marks, and outlined type for the printer.

2. Generation pipeline

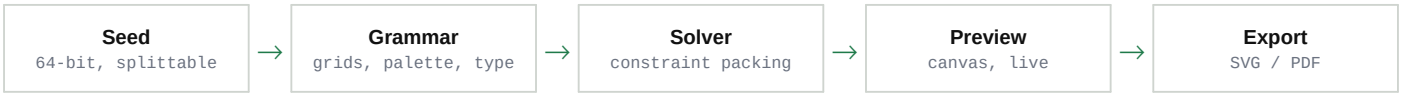

Figure 1 — Preview and export share one layout pass; the canvas renderer and the vector emitter consume the same resolved scene graph.

The generator draws from a splittable PRNG rather than a single stream. Each subsystem — grid subdivision, palette selection, type sizing, ornament placement — receives its own derived stream, so adding a decision to one subsystem does not shift every downstream draw and invalidate previously approved seeds.

3. System layers

Grammar	Declarative TypeScript DSL for grid rules, palette sets, type scales, and ornament families.
Sampler	Splittable seeded PRNG with per-subsystem streams and weighted choice helpers.
Layout solver	Packs content into the sampled grid, enforcing margin, optical balance, and collision rules.
Scene graph	Resolved, renderer-agnostic tree of shapes, text runs, and colour references.
Renderers	Canvas for interactive preview; SVG and PDF emitters for final artwork.

4. Grammar primitives

PRIMITIVE	DECLARES	SAMPLED AS
grid	Column and row counts, gutters, margin ratios.	One subdivision per poster.
palette	Named colour sets with role constraints.	Set choice, then per-role assignment.
typeScale	Ratio, base size, permitted weights.	Scale step per text role.
ornament	Shape family, density, placement zones.	Count and positions within zones.

5. Grammar definition

```
export const poster = defineGrammar({
  version: "1.6",                 // pinned into every export
  canvas: { w: 594, h: 841, unit: "mm", bleed: 3 },   // A1
  grid:   { cols: [6, 9, 12], rows: [8, 12], gutter: "1.5%", margin: "6%" },
  palette: [
    { name: "acid",  colors: ["#0B0B0B", "#F2F0E9", "#C6F24E"] },
    { name: "oxide", colors: ["#161310", "#EDE6DA", "#C2451F"] },
  ],
  typeScale: { base: 14, ratio: [1.25, 1.333, 1.5], weights: [400, 700] },
  rules: [
    minContrast(4.5),              // title against its backing field
    keepClear("title", "ornament", 8),
    balance({ axis: "optical", tolerance: 0.12 }),
  ],
});

render(poster, { seed: "0x9F3C1A62", target: "pdf" });
```

Listing 1 — The same grammar and seed always produce the same poster, on any machine.

6. Reproducibility guarantees

- **Pinned grammar.** Exports embed the grammar version and seed in XMP metadata. A grammar change bumps the version; old versions stay resolvable so archived posters still regenerate.
- **Stream isolation.** Subsystem PRNG streams are derived from the root seed by name, not by draw order, so adding an ornament rule cannot alter the grid or palette of an existing seed.
- **Font locking.** Text metrics are resolved against a font hash recorded at export time; a font update that would reflow the layout is flagged rather than silently applied.

7. Print output

PROPERTY	SETTING	NOTES
Colour	PDF/X-4, CMYK or spot	ICC profile embedded; RGB retained for digital exports.
Bleed and marks	3 mm bleed, crop marks	Configured per canvas, not per export.
Type	Outlined on export	Avoids font dependency at the printer.
Raster elements	Rejected above 0 dpi	The pipeline is vector-only by design.

8. Workflow notes

- A contact-sheet mode renders 48 seeds at thumbnail scale in one pass, which is how most selection actually happens; picked seeds promote to full resolution individually.
- Grammars live in the repository and are reviewed like code. Studio output stays consistent because the constraints are versioned artifacts rather than habits.
- Export is deterministic and headless, so the same command runs in CI to regenerate an entire campaign after a palette correction.

9. Roadmap

RELEASE	PLANNED WORK	TARGET
1.7	Animated variants exporting to SVG SMIL and MP4 for social formats.	Q4 2026
1.8	Constraint debugger showing which rule rejected a candidate layout.	Q1 2027
2.0	Multi-page systems: covers, spreads, and shared grammar across a series.	Q2 2027